KNECK OFFICIAL

Chemistry — Paper 2

233/2 • ORIGINAL REVISION SET 01 • 2 hours • 80 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all seven structured questions. Values: H=1,C=12,O=16,Na=23,Mg=24,Cl=35.5,Ca=40,Cu=64; molar gas volume 24 dm³/mol. Show all working.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

STRUCTURED QUESTIONS

All questions compulsory.

1. Periodic table: A has electron arrangement 2.8.1, B 2.8.7, C 2.8.8. (a) Identify A,B,C. (3) (b) Explain the bond between A and B. (4) (c) Explain why C is unreactive. (2) (d) Compare the atomic radii of A and B. (3) [12 marks]

2. 25.0 cm³ of 0.100 mol/dm³ HCl neutralises 20.0 cm³ NaOH. (a) Write equation. (2) (b) Calculate moles acid and concentration alkali. (4) (c) Calculate mass NaOH in 250 cm³ alkali. (2) (d) Explain rinsing burette with acid and conical flask with distilled water. (4) [12 marks]

3. A metal carbonate sample of 5.0 g gives 0.96 dm³ CO2 with excess acid. The pure carbonate is CaCO3. (a) Write balanced equation with HCl. (2) (b) Calculate moles CO2 and mass CaCO3. (4) (c) Calculate percentage purity. (2) (d) Describe a test for CO2 and explain observation. (4) [12 marks]

4. For equilibrium N2(g)+3H2(g) ⇌ 2NH3(g), forward reaction is exothermic. (a) Predict effects of increased pressure and temperature on ammonia yield; explain. (6) (b) Explain use of an iron catalyst. (2) (c) Explain recycling of unreacted gases. (2) [10 marks]

5. Magnesium displaces copper from copper(II) sulfate. A cell uses Zn/Zn²+ (E°=-0.76 V) and Cu²+/Cu (E°=+0.34 V). (a) Write ionic Mg reaction. (2) (b) Identify anode/cathode and write half equations. (4) (c) Find standard cell voltage. (2) (d) State salt bridge function. (2) [10 marks]

6. Consider ethane, ethene and ethanol. (a) Give molecular formulae and classify each. (3) (b) Distinguish ethane from ethene chemically. (3) (c) Describe fermentation conditions and write equation from glucose. (4) (d) Name oxidation product of ethanol under reflux with excess acidified oxidant. (2) [12 marks]

7. A reaction releases 5.0 kJ into 100 g water and raises its temperature. Water heat capacity = 4.2 J/g/°C. (a) Calculate temperature rise. (3) (b) State two causes of experimental heat loss/error and

improvements. (4) (c) Define activation energy and distinguish exothermic from endothermic change. (5) [12 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

STRUCTURED QUESTIONS

1. (12 marks) A sodium, B chlorine, C argon (3). One electron transferred A→B, A+ and B-, electrostatic attraction (4). Full outer shell gives stability (2). Na larger: same occupied shells but increasing nuclear attraction across period pulls electrons closer in Cl (3).

2. (12 marks) HCl+NaOH→NaCl+H2O (2). Acid n=.00250; alkali n=.00250; c=.125 mol/dm³ (4). Mr=40; mass=.125×.250×40=1.25 g (2). Burette rinse prevents dilution (2); flask water changes volume but not moles of measured alkali (2).

3. (12 marks) CaCO3+2HCl→CaCl2+H2O+CO2 (2). n=.96/24=.04 mol; mass=4.0 g (4). Purity=80% (2). Bubble into limewater; turns milky; Ca(OH)2+CO2→CaCO3(s)+H2O; precipitate causes cloudiness (4).

4. (10 marks) Pressure raises yield because 4 gas moles become 2 (3). Temperature increase lowers equilibrium yield because reverse endothermic reaction favoured (3). Catalyst increases rate/lower activation energy but does not change equilibrium position (2). Recycling improves overall conversion and reduces waste (2).

5. (10 marks) Mg+Cu²+→Mg²++Cu (2). Zinc anode Zn→Zn²++2e-; copper cathode Cu²++2e-→Cu (4). E°cell=.34-(-.76)=1.10 V (2). Allows ion migration/completes circuit and maintains electrical neutrality (2).

6. (12 marks) C2H6 alkane, C2H4 alkene, C2H5OH alcohol (3; require paired entries). Bromine water decolourised by ethene, not ethane in absence of UV; addition across C=C (3). Yeast, warm ~30–35°C, anaerobic; C6H12O6→2C2H5OH+2CO2 (4). Ethanoic acid CH3COOH (2).

7. (12 marks) ΔT=5000/(100×4.2)=11.90°C (3). Heat to air: insulate/lid; heat to container: calorimeter correction; incomplete transfer: stir/rapid record; any 2 paired limitations and improvements (4). Activation energy minimum collision energy needed for reaction (2); exothermic transfers heat out ΔH<0 (2); endothermic heat absorbed ΔH>0 (1).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf